Explainable Machine Learning and a Grounded Policy Retrieval Prototype for Food Inflation Forecasting and Economic Early Warning in Bangladesh

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Keywords— *Food inflation forecasting, Explainable machine learning, Economic early warning, Conformal prediction, Policy retrieval.*

1. Research Area

Applied machine learning and computational macroeconomics (time-series forecasting). Research question: can gradient-boosted models trained on publication-lag-aligned daily market data and monthly macro indicators forecast Bangladesh's food inflation better than linear and naive-persistence baselines, and are the resulting early-warning signals reliable enough to be useful to policymakers?

2. Goals/Objectives

Official food CPI and reserve figures from the Bangladesh Bureau of Statistics (BBS) and Bangladesh Bank appear around Day 7–10 of the following month. This delay is modest, but during shock regimes (currency depreciation, commodity spikes) it leaves monetary and procurement authorities without a timely signal. We therefore test, as an honest benchmark rather than a claim of superiority: (i) whether gradient boosting improves one-month and three-month-ahead food inflation forecasts over linear regression and naive persistence; (ii) how well rolling conformal prediction calibrates uncertainty bands for retail staple prices; (iii) whether unsupervised anomaly detection gives advance warning of macroeconomic stress episodes without excessive false alarms; and (iv) how a policy retrieval pipeline prototype performs on domain queries.

3. Methodology

Data. A daily panel (January 2010–September 2026, 6,117 observations, 41 indicators) is combined with a monthly panel of official Bangladesh Bank/BBS series (201 months, no interpolation). Release schedules (BBS CPI on Day 10 of M+1, Bangladesh Bank reserves on Day 7 of M+1, WFP retail surveys on Day 15 of M) are enforced by step-function forward fills; daily international series (Brent crude, CBOT wheat, US Dollar Index, natural gas, gold) enter contemporaneously. Known limitation: annual World Bank indicators are step-filled by calendar year and are not lagged to their release dates, which introduces look-ahead bias that we have not yet removed. **Forecasting.** XGBoost [1] is trained on chronological splits (train ≤ 2021, validation 2022–2023, test 2024–2026) and compared with Ridge regression, Random Forest and naive persistence using Harvey-Leybourne-Newbold corrected Diebold-Mariano (DM) tests [3]; feature attributions use TreeSHAP. **Uncertainty.** For coarse rice, lentils and wheat flour, rolling walk-forward conformal prediction (730-day calibration window) targets 80% intervals; time-series dependence means no formal coverage guarantee holds. **Early warning.** An Isolation Forest [2] is fitted only on pre-2022 data, combined with a resetting CUSUM filter, and evaluated out-of-sample (2022–2026) against documented stress episodes. **Retrieval.** A ChromaDB dense-retrieval prototype indexes four policy summaries (Bangladesh Bank, BBS, IMF, TCB).

4. Results and Analysis

Forecasting. On the monthly test set (2024–2026, n ≈ 33), XGBoost lowers RMSE relative to Ridge by 20.0% at t+1 and 32.5% at t+3, but neither difference is significant at the 5% level (DM p = 0.068, 0.058). XGBoost does not beat naive persistence (t+1: 1.171% vs. 1.160%) or Random Forest (Table I). In daily 30-day-ahead forecasting it clearly beats Ridge (DM = 3.74, p < 0.001) yet stays slightly behind persistence (1.158% vs. 1.137%); because the target is a step-filled monthly series, the effective sample is small and this p-value is optimistic. TreeSHAP ranks reserve adequacy (months of imports, 19.3%), remittances (15.9%) and headline-inflation persistence (15.4%) highest, whereas Brent crude (1.4%) and wheat (1.2%) contribute little; since the top feature is an annual series, this ranking is itself affected by the look-ahead issue above.

Horizon	Naive	Ridge	Random Forest	XGBoost (vs. Ridge)	DM vs. Ridge (p)
t+1 (1 month)	1.160%	1.464%	1.087%	1.171% (-20.0%)	1.82 (0.068)
t+3 (3 months)	1.757%	2.578%	1.725%	1.740% (-32.5%)	1.90 (0.058)
30-day daily	1.137%	2.457%	—	**1.158% (-52.9%)**	**3.74 (<0.001)**

Table I: Out-of-sample RMSE and Diebold-Mariano tests (lowest RMSE per row underlined).

Uncertainty and early warning. Over 1,368 out-of-sample days, conformal intervals reached 83.6% empirical coverage for coarse rice but only 76.0% (lentils) and 75.2% (flour) against the 80% target. The anomaly detector flagged both out-of-sample episodes (2022 FX drain, 2024 political shock) 80 and 90 days before onset, but it also flagged 74% of out-of-sample days. Most of these days lie outside the pre-2022 training range (e.g., USD/BDT), and the lead time is bounded by a 90-day search window, so it should be read as a distribution-shift signal. Precision is 56.5% against a 42% crisis-day

base rate (F1 0.72 vs. 0.59 for always alarming), so alert-persistence filtering is required before any operational use. **Retrieval.** The prototype reached 82.0% mean keyword coverage on 10 queries over four policy summaries; LLM generation and hallucination have not yet been evaluated, so no grounding claim is made.

Open issues. Calendar-year look-ahead in annual indicators; a short test period (≈33 months) with two crisis episodes; hand-labelled crisis windows; incomplete conformal calibration for lentils and flour; and absence of a generation-level retrieval evaluation. Code and data are released openly; corrected results will be versioned in the repository.

5. References

[1] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 785–794.

[2] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, 2008, pp. 413–422.

[3] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *J. Business & Economic Statistics*, vol. 13, no. 3, pp. 253–263, 1995.